An Integrated AI-Powered Decision Support System for Sustainable Farming Practices in India

Quantified Sustainability Scoring, Multi-Stakeholder Analysis, and Government Scheme Matching

1st Gaurav Singh
Department of Computer Science & Engineering
CGC College of Engineering, Landran
Mohali, Punjab 140307, India

2nd Sarang Ahlawat
Department of Computer Science & Engineering
CGC College of Engineering, Landran
Mohali, Punjab 140307, India

3rd Rujul Gupta
Department of Computer Science & Engineering
CGC College of Engineering, Landran
Mohali, Punjab 140307, India

4th Ashpinder Kaur
Assistant Professor
Chandigarh Group of Colleges, Landran
Mohali, Punjab, India

***Abstract*—The agriculture sector constitutes a vital base of the Indian economy, accounting for around 18–20% of the GDP of the country and employing more than half of the total labor force in the nation, but still faces issues like soil deterioration, heavy dependency on chemicals, poor participation in governmental schemes, insufficient market connections, and production threats due to climatic factors [2]. This paper proposes a Decision Support System with AI assistance for sustainable agriculture which uses multi-source data related to the government and environment, stakeholder profiling, sustainability measurement, and explanation of recommendations. The proposed system makes use of a novel Quantified Sustainability Scoring Model which evaluates farms' sustainability on the parameters of soil health, water management, reduced chemical consumption, biodiversity preservation, and economic robustness. Based on the Punjab state-centric datasets, the framework aims at providing scheme matching, sustainability recommendations, and future forecasts on the healthiness of farmland. This decision support system is designed to be used by farmers, government officers, and researchers.**

***Index Terms*—AI Decision Support System, Sustainable Agriculture, QSSM, Scheme Matching, Explainable AI, Indian Agriculture**

I. Introduction

The agriculture industry in India finds itself at an important juncture where climate change issues, degradation of farmland, and poor information integration are making conventional advisory services and government aid programs ineffective [2]. Even though there are around 40 different types of agriculture and social assistance programs, many smaller and marginal farmers are either unaware of them or fail to benefit from them owing to various complications and lack of contextualized advice [9]. There are various sustainable farming methods, such as crop rotation, agroforestry, integrated nutrient management, integrated pest management, and irrigation management, which are proven effective and essential for productive agriculture practices; however, they have failed to reach the desired level of adoption owing to the lack of farm-specific suggestions.

For instance, a farmer in the state of Punjab might be aware of the concept of drip irrigation but might not be sure about whether drip irrigation would be financially feasible on a 2.5-acre plot, whether the applicable subsidy is available, or whether local conditions merit such a shift to drip irrigation technology [5] [12]. Hence, the problem goes deeper than just being a lack of knowledge. The core problem lies in the fact that farmers are missing out on a holistic decision-making capability. This is what the development of an AI-based DSS will address [6]. This paper proposes a model for the design of an integrated AI-DSS. The key aspects of the proposed design and the functionalities associated with it are taken directly from the project synopsis and are based on four layers, seven stakeholders, and farm sustainability evaluation in India.

TABLE I
Key Problem Areas in Indian Agricultural Decision-Making

Problem Area	Impact
Data Fragmentation	Data in agriculture continues to be stored in more than 30 government websites with no common access interface.
Generic Advice	Many current advice-giving programs give generic recommendations and do not consider local conditions.
Scheme Utilization Gap	Many farmers who qualify for gov- ernment programs are unaware because of knowledge constraints.
Stakeholder Exclusion	Tenant farmers, women farmers, FPOs, and laborers are usually left out of decision-making processes.
No Sustainability Metric	There is no common score for sustainability progress at the farm level.
Language Accessibility	Access Problem Information regarding agriculture is gen- erally inaccessible to people using regio- nal languages.
Long-Term Forecasting Gap	Current systems mostly forecast seasonal yield and not the health of the land in the long run.

II. Related Work

Agricultural Decision Support Systems have progressed from being purely expert systems like COMAX and GOSSYM to become sophisticated crop models like DSSAT and APSIM [3] [4]. Although these systems provide powerful capabilities for crop modeling, they are mainly for scientific research purposes, without much consideration of usability by farmers in rural India [3]. Agri-information platforms in India provide information but are largely non-prescriptive in nature [5].

A number of recent studies on the use of AI in agriculture have proved very efficient in predicting the yield of crops, detecting diseases in plants, optimizing irrigation, and analyzing the soil [1] [7]. Nonetheless, many of these studies tend to be centered on productivity or efficiency gains, while paying little attention to the results achieved in terms of sustainability and stakeholders' involvement [9] [12].

These include a number of unsolved gaps such as lack of multi-portal integration, modeling with the involvement of multiple stakeholders, inadequate sustainability measurement, failure to forecast environmental impacts over the long run, and limited explainability of AI recommendations.

TABLE II
Research Gap Analysis

Existing Research Covers	This Work Addresses
Crop production forecast	Farm-level sustainability scoring
Unilateral stakeholder advice	Complete multistakeholder modelling
General national analysis	Customized farm, soil-based advice
Isolated government data sources	Single multichannel pipeline
English-only interfaces	Regional language support
Season-based predictions	Land health forecasts of 3-10 years
Scheme information dissemination	Automated scheme eligibility matching
Black-box prediction outputs	AI predictions with SHAP reasoning

III. Proposed System Architecture

The AI-based DSS has been structured in four coherent layers that include data collection and integration, stakeholder and land assessment, sustainable farming practices intelligence, and AI-supported decision-making support improvement. The above-mentioned layers ensure the conversion of disintegrated data into meaningful recommendations and scheme-matching insights.

Data ingestion layer is responsible for bringing together the government portal, surveys, environment data, and market feed in a normalizing pipeline of analytics. Stakeholder and land analysis layer creates profiles for farmers, tenant farmers, women farmers, FPO, labor, officer, and researcher profiles. Sustainable farming intelligence layer is responsible for creating an organized knowledge base of sustainability practices along with sustainability scores. Finally, a recommendation layer brings out farm-based suggestions based on previous layers' output.

TABLE III
Four-Layer AI-DSS Architecture

Layer	Name	Function
1	Data Ingestion & Aggregation	Ingests, cleans, harmonizes, and stores mul- tisource governmental, survey, market, and envi- ronmental datasets.
2	Stakeholder & Land Analysis	Generates stakeholder profiles and land unit profiles based on socioeconomic indicators and land health indices.
3	Sustainable Practice Intelligence	Assesses practices, analyzes suitability, and calculates sustainability metrics.
4	AI Decision Support Engine	Generates explainable recommendations, scheme matching, policy suggestions, and fore- casts.

A. Layer 1: Data Ingestion and Aggregation

The integration layer comprises datasets related to PM-KISAN, eNAM, Soil Health Card, PMFBY, PMKSY, Agri-Census, NSSO, IMD, ISRO Bhuvan, and State Agriculture Departments. The data pipeline will perform tasks like schema normalization, missing value imputation, geo-alignment at the level of districts and villages, and time-series alignment for season-wise and year-wise analysis.

B. Layer 2: Stakeholder and Land Profiling

In this level, the stakeholders will be profiled in an organized way and the land health will also be profiled. The stakeholders will include not only the owners of the land, the farmers, but also the tenant farmers, women farmers, laborers, cooperatives, and the officers.

TABLE IV
Modelling Of Stackholder's type in the System

Stakeholder Type	Representative Variables
Small Farmers	Land area, crop history, soil composition, scheme participation, indebtedness, literacy level
Tenant Farmers	Land rights, likelihood of exclusion, insur- ance and credit access
Women Farmers	Ownership, association with SHGs, scheme access based on gender
Farm Laborers	Dependence on wages, seasonal labor, mi- gration trends
FPOs / Cooperatives	Membership, market connections, storage facili- ties, crop diversification
District Officers	Scheme coverage, subsidy disbursement, training outcomes
Researchers	Data export requirements, publication connectivi- ty, clinical trials

C. Layer 3: Sustainable Farming Practice Intelligence

The heart of the system from a scientific standpoint is the knowledge base that relates sustainable farming practices to their respective applicability criteria, transition criteria, outcome criteria, and government programs. In this layer, the Quantified Sustainability Scoring Model (QSSM) is introduced as an index measuring sustainability on a farm scale.

$$QSSM = 0.25S_h + 0.20W_s + 0.20C_r + 0.20B_e + 0.15E_r \quad (1)$$

In this context, S_h stands for Soil Health, W_s for Water Stewardship, C_r for Chemical Input Reduction, B_e for Biodiversity and Ecosystem Resilience, and E_r for Economic Resilience. The criteria and weights are derived from the

TABLE V
QSSM DIMENSIONS AND WEIGHTS

Dimension	Key Indicators	Weight
Soil Health	Organic carbon, pH, NPK, micro- nutrient content	25
Water Management	Water application efficiency, groundwater reliance	20
Chemical Usage Reduction	Pesticides, fertilizer dependency, integrated pest management	20
Biodiversity & Ecosystem	Balance Crop diversity, agrofor- estry practices, ecological balance	20
Economic Sustainability	Diversification of income sources, market access	15

D. Layer 4: AI Decision Support and Recommendation

This final layer integrates all the elements mentioned above into contextual results, which include farm state analysis, identification of the best sustainable practices, recommendations on government schemes to be applied, identification of gaps in policies, and land health forecast. The conclusion clearly highlights the importance of explainability for this project and suggests the use of SHAP approach.

IV. METHODOLOGY

The research adopts a Design Science Research approach that is ideal for developing and assessing information systems artifacts. The design science research process comprises six phases, including identifying the problem, defining objectives, designing the system, demonstrating it, evaluating it, and communicating its results.

The primary data will be collected using questionnaires from farmers, district agriculture officers, FPO members, and researchers, whereas the secondary data will be retrieved from government websites, questionnaires, weather agencies, and agricultural research institutions. The research design, thus, combines system design and empirical validation, consistent with the research paper's two-fold research objective.

TABLE VI
RESEARCH METHODOLOGY PHASES

Phase	Name	Key Activity
1	Problem Identification	Literature review, survey of the field, portal analy- sis
2	Objective Definition	Dimensions of QSSM, criteria for suc- cess, expert advice
3	Design & Development	Architecture, data flow, machine learn- ing and natural language processing components
4	Demonstration	Piloting with actual farm data
5	Evaluation	Precision, accuracy, usability, fairness anal- ysis
6	Communication	Research paper, technical report, policy memorandum

A. AI and Machine Learning Techniques

Machine learning, natural language processing, clustering, optimization, rule-based, and explainable AI techniques will be used in the system design process. The use of Random Forest and XGBoost is considered for risk prediction and scoring, the use of collaborative filtering for recommendation ranking, IndicBERT or other multi-language models for regional language understanding, LSTM for long-term forecasting, NSGA-II for compromise optimization, and SHAP for explanation of recommendations.

TABLE VII
AI AND MACHINE LEARNING TECHNIQUES

Technique	Application	Why Used
Random Forest / XGBoost	QSSM scoring, crop risk	Performs effectively on agriculture datasets with diverse information
Collaborative Filtering	Practice ranking	Recommendation system Makes use of farm profiles that are alike
IndicBERT / NLP Models	Punjabi/Hindi processing	Supports regional language interaction
LSTM	Long-term forecasting	Works well with temporal information
NSGA-II	Tradeoff analysis	Provides a balance between sustainability and practicality
K-Means / DBSCAN	Farmer clustering	Classifies farmers based on common characteristics
Rule-Based System	Scheme matching	Employs deterministic eligibility criteria
SHAP	Explainability	Provides explanations for recommendations

B. Technology Stack

The suggested stack will involve Python, Pandas, NumPy, Apache Airflow, FastAPI, PostgreSQL, Redis, PostGIS, Geopandas, React Native/PWA interfaces, React.js dashboards, and cloud services like AWS or Azure. Such choices correspond to the needs of research-based analytics along with user-friendly interfaces designed for various stakeholders.

TABLE VIII
SYSTEM TECHNOLOGY STACK

Layer	Technology	Purpose
Data Pipeline	Python, Pandas, NumPy, Airflow	ETL and preprocessing
AI / ML Engine	Scikit-learn, PyTorch, HuggingFace	Model training and NLP
Backend API	FastAPI, PostgreSQL, Redis	APIs, storage, caching
Geospatial Layer	PostGIS, GeoPandas, Folium	Spatial analysis and mapping
Farmer Interface	React Native / PWA	Mobile-first user access
Officer Dashboard	React.js, Recharts, Leaflet	Interactive officer visualizations
Research Dashboard	Streamlit / Dash / Plotly	Analytical and policy reporting
Security	OAuth 2.0, JWT, AES-256	Authentication and data protection

V. EXPECTED OUTCOMES

These outputs include a functional decision support system using AI, a multilingual application for farmers, an online portal for government officers, and a research and policy dashboard. The expected outputs include the validation of the QSSM tool, a sophisticated scheme-matching tool, and publishable findings regarding sustainability scoring, scheme implementation gaps, and collaborative decision making for agriculture stakeholders.

The expected social impacts are access to schemes, increased adoption of sustainable practices, enhanced farmer livelihood, decreased soil degradation, and informed policies. This is in line with the vision of the project as an initiative that goes beyond being just a software solution but a sustainable agricultural transformation process.

TABLE IX
EXPECTED OUTCOMES AND IMPACT

Impact Area	Expected Outcome
Scheme Participation	20–35% rise in scheme participation among farmers in the profiles
Practice Implementation	Evidence of QSSM improvement through individual transition strategies
Income Stability	Increased income stability through diversifica- tion and scheme participation
Soil Quality	Soil degradation reduced; organic carbon improved over time periods
Scientific Community	Reusable datasets, methods, and publishable models

VI. DISCUSSION

One of the most valuable contributions of this architecture is its ability to advance the current practice of agricultural AI to incorporate not just prediction but also integrated decision support systems. Rather than being solely centered around crop yield, the proposed architecture takes into account sustainability, accessibility in policies, the conditions of the land, diversity in stakeholders, and cost-effectiveness. Such an approach is necessary because the inability to adopt sustainable farming practices is more frequently due to friction than a lack of scientific evidence.

Another key contribution is related to the issue of explainability. The financial risks involved with agricultural decisions require more than good recommendations. The project's focus on explainable decision support via SHAP-like explanations and reasoning is expected to increase trust and ensure officer oversight. Lastly, the design is scalable; the architectural synopsis specifically mentions voice-enabled devices, Internet-of-Things sensor integration, federated learning, and carbon credits.

VII. CONCLUSION

This research aims to develop a Decision Support System based on Artificial Intelligence that will consolidate information from multiple sources, build stakeholder models, incorporate QSSM ratings, and provide explainable solutions for the identified challenges in India's agricultural advisory services system.

The project will not only bridge the gap between implementation and research but also contribute to making agricultural decisions in India more inclusive, measurable, and evidence-based.

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REFERENCES

[1] S. Khaki and L. Wang, "Crop yield prediction using deep neural networks," *Frontiers in Plant Science*, vol. 10, p. 621, 2019.

[2] Ministry of Agriculture and Farmers Welfare, "Annual Report 2022–23," Government of India, 2023.

[3] J. W. Jones et al., "The DSSAT cropping system model," *European Journal of Agronomy*, vol. 18, no. 3–4, pp. 235–265, 2003.

[4] B. A. Keating et al., "An overview of APSIM, a model designed for farming systems simulation," *European Journal of Agronomy*, vol. 18, no. 3–4, pp. 267–288, 2003.

[5] R. Kumar and M. P. Singh, "Role of information and communication technology in Indian agriculture: A review," *International Journal of Current Microbiology and Applied Sciences*, vol. 7, no. 8, pp. 1527–1535, 2018.

[6] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt, "Big data in smart farming – A review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017.

[7] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep learning in agriculture: A survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[8] R. Bhattacharyya et al., "Soil degradation in India: Challenges and potential solutions," *Sustainability*, vol. 7, no. 4, pp. 3528–3570, 2015.

[9] IFPRI, "2019 Global Food Policy Report," International Food Policy Research Institute, 2019.

[10] S. Palekar, "Zero Budget Spiritual Farming," Zero Budget Natural Farming Research, Development and Extension Movement, 2006.

[11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, 2017, pp. 4765–4774.

[12] R. Kochhar et al., "Digital agriculture in India: Status, challenges and opportunities," NITI Aayog Discussion Paper, Government of India, 2021.